Jack A. Logan

Research Profile

Theoretical and computational physicist with a PhD and 9+ years of experience working at the intersection of statistical mechanics, geometry, and machine learning. My research focuses on emergent structure, representation learning, and physically grounded modeling, with applications spanning soft matter, polymers, proteins, and interpretable neural networks. I work across theory, simulation, and computation to develop models that reveal underlying mechanisms and structure in complex physical systems.

Selected Projects & Research

Independent Research

Surrogate Neural Network Force Field
PyTorch | Molecular Dynamics | Statistical Mechanics

- Built PyTorch **neural network surrogate force field for Lennard-Jones systems** using radial basis function features and central-force structure.
- Designed full training pipeline with **physics-informed representations** and force reconstruction.
- Benchmarked against MD simulations and **evaluated using physically meaningful observables** including force consistency and radial distribution functions demonstrating accurate recovery of physical behavior.

Participation Ensembles: A Statistical Framework for Interpretable Neural Representations
PyTorch | Interpretability | Entropy | Regularizers | (In Progress)

- Developed **statistical framework for neural representations** using entropy-based diagnostics and interaction measures.
- Proposed regularization methods to **control feature mixing and structure** in learned representations.

Multi-Agent Communication: Guide and Explorer
PyTorch | Gymnasium | PettingZoo | Stable-Baselines3S | (In Progress)

- Developing a toy environment where an explorer must traverse a hazardous grid while a guide communicates safe paths through a noisy channel.
- Designed to test **coordination, emergent communication, and collective problem solving**.

Emergent Dihedral Symmetry in a 1-Layer Transformer
PyTorch | Group Theory | Interpretability

- **Trained and analyzed a 1-layer Transformer** to learn dihedral group D3 operations.
- Identified failure of compositional closure via mechanistic interpretability analysis.

Jamming On Curved Manifolds
C++ | Constrained Molecular Dynamics | Geometric Frustration

- Designed and implemented a C++ **MD engine for simulating polydisperse particles constrained to curved surfaces** with variable radii (spheres, tori, cylinders, and planes).
- Built **custom meshing algorithms** to measure packing fractions and study **jamming transitions on 2D curved manifolds**.

Academic Research (PhD/Postdoc)

Structural Properties of Folded Proteins
C++ | Python | Molecular Dynamics

- Developed C++ **molecular dynamics framework** for coarse-grained protein folding.

- Ran **large ensembles of simulations** (thousands of configurations) on HPC systems to study emergent structural features.
- Connected **interaction rules to observed folding behavior** through statistical analysis.

SymBOPs: Symmetrized Bond Order Parameters
Group Theory | Tensors | Symmetry Representations | Computational Validation
- Developed symmetry-specific bond order parameters **grounded in group theory**.
- **Validated on experimental and simulated self-assembled systems**, identifying symmetry-defined domains.

Geometric and Topological Entropies of Sphere Packing
Statistical Mechanics | Monte Carlo
- Developed a **statistical-mechanical framework to quantify entropy** in random sphere packings that generalizes concepts of granular and glassy configurational entropies for the case of non-jammed systems.
- Designed **custom Monte Carlo algorithm** to generate and sample constrained packing ensembles.
- Used **computational experiments to build constrained sphere packings** and relate packing geometry and geometric and topological entropy.

Compact Interaction Potential for van der Waals Nanorods
Analytical Framework | Asymptotics
- Derived compact **analytical expressions for van der Waals interactions** between nanorods in arbitrary relative configurations.
- Enabled efficient **modeling of anisotropic self-assembly** without full-scale particle simulations.
- Provided immediately **usable interaction models for computational studies** of nanorod assemblies.

Research Tools

Programming & HPC: Python, C++, CUDA, parallel computing on HPC clusters
Modeling & Simulation: Molecular Dynamics, Monte Carlo methods
Machine Learning: PyTorch, neural networks, physics-informed ML, mechanistic interpretability
Data & Collaboration: statistical analysis, optimization, Git/GitHub

Professional Experience

Yale University, New Haven, CT
Postdoctoral Fellow, Mechanical Engineering & Materials Science
Advisor: Corey O'Hern
2022-2025

Brookhaven National Laboratory, Upton, NY
Research Assistant, Center for Functional Nanomaterials
Advisor: Alexei V. Tkachenko
2016-2022

Selected Publications

- Logan, J.A., et al. (2025) *Effect of stereochemical constraints on the structural properties of folded proteins*. Physical Review E, *Editors' Suggestion.*
- Logan, J.A., et al. (2023). *Symmetry-specific characterization of bond orientation order in DNA-assembled nanoparticle lattices*. Journal of Chemical Physics, *Editors' Pick.*

- Logan, J.A., Tkachenko, A.V., (2022). *Geometric and Topological Entropies of Sphere Packing*. Physical Review E.
- Mushnoori, S., Logan, J.A., et al. (2022). *Controlling morphology in hybrid isotropic/patchy particle assemblies*. Journal of Chemical Physics, *Editors' Pick*.
- Logan, J.A., et al. (2022). *Symmetry-specific orientational order parameters for complex structures*. Journal of Chemical Physics.
- Logan, J.A., Tkachenko, A.V., (2018). *Compact interaction potential for van der Waals nanorods*. Physical Review E.

Education

Stony Brook University, Stony Brook, NY
PhD in Physics, *2022*
Advisor: Alexei Tkachenko
Dissertation: *Geometric and Topological Aspects of Self-Assembly: from spheres and rods to designer particles*

Stony Brook University, Stony Brook, NY
Master of Arts in Physics

Long Island University, Brookville, NY
Master of Science in Applied Mathematics
Summa Cum Laude

Leadership & Awards

- Mentor, student Summer projects, Yale University (2024)
- GSNP Postdoctoral Speaker Award Finalist, APS March Meeting (2024)
- Team Leader, Protein Folding Simulation Project, Yale University (2022-2025)
- PEB Travel Fund Award, Yale University (2023)
- Postdoctoral Scholars Travel Fund Award (2023)
- Coordinated interdisciplinary collaborations to computationally and experimentally validate the SymBOPs theoretical framework (2021)
- Jonathan Kaufman Student Excellence Prize (2019)